Stage 1: Cleaning and Preprocessing Planet Data

In [1]:
```
# This code was created by me using lecture videos from the Data Science course
#and Database and Advanced Data Techniques and previous projects in Data Science li


#load libraries
import pandas as pd
import numpy as np
```

Planet Column Data

In [2]:
```
#load full column data file
column_data=pd.read_csv(r"C:\Users\maybe\OneDrive\Desktop\Databases and Advanced Te

#check first rows to verify it loaded correctly
print(column_data.head(1))
```

```
   column_id abbrev_name  full_name
0          1   planet_id  Planet ID
```

In [3]:
```
#### Find critical columns in column data
critical_column_data=['pl_name', 'hostname', 'pl_letter', 'sy_snum', 'sy_pnum',
                       'sy_mnum','discoverymethod', 'disc_year', 'disc_facility',
                       'pl_orbper','pl_orbsmax', 'pl_angsep' , 'pl_rade', 'pl_bma
                      'pl_dens', 'pl_orbeccen', 'pl_insol', 'pl_eqt', 'pl_orbincl
                      'pl_trandep', 'pl_trandur', 'pl_rvamp', 'st_spectype', 'st_
                      'st_met', 'st_lum', 'st_logg', 'st_age', 'st_dens', 'st_vsi
                      'st_radv', 'ra', 'dec', 'sy_dist']

new_column_data = column_data[column_data['abbrev_name'].isin(critical_column_data)

#new_column_data = column_data[critical_column_data]
print(new_column_data)
```

```
     column_id      abbrev_name                                  full_name
1            2          pl_name                                Planet Name
2            3         hostname                                  Host Name
3            4        pl_letter                              Planet Letter
9           10          sy_snum                            Number of Stars
10          11          sy_pnum                          Number of Planets
11          12          sy_mnum                            Number of Moons
13          14  discoverymethod                           Discovery Method
14          15        disc_year                             Discovery Year
18          19    disc_facility                         Discovery Facility
32          33        pl_orbper                      Orbital Period [days]
37          38       pl_orbsmax                  Orbit Semi-Major Axis [au]
42          43       pl_angsep                    Angular Separation [mas]
47          48          pl_rade                Planet Radius [Earth Radius]
57          58        pl_bmasse   Planet Mass or Mass*sin(i) [Earth Mass]
68          69          pl_dens                   Planet Density [g/cm**3]
73          74       pl_orbeccen                               Eccentricity
78          79         pl_insol                Insolation Flux [Earth Flux]
83          84           pl_eqt               Equilibrium Temperature [K]
88          89       pl_orbincl                          Inclination [deg]
93          94       pl_tranmid                    Transit Midpoint [days]
105        106       pl_trandep                          Transit Depth [%]
110        111       pl_trandur                   Transit Duration [hours]
141        142         pl_rvamp             Radial Velocity Amplitude [m/s]
156        157      st_spectype                              Spectral Type
163        164           st_rad                Stellar Radius [Solar Radius]
168        169          st_mass                  Stellar Mass [Solar mass]
173        174           st_met                  Stellar Metallicity [dex]
179        180           st_lum             Stellar Luminosity [log(Solar)]
184        185          st_logg  Stellar Surface Gravity [log10(cm/s**2)]
189        190           st_age                          Stellar Age [Gyr]
194        195          st_dens                  Stellar Density [g/cm**3]
199        200          st_vsin          Stellar Rotational Velocity [km/s]
204        205          st_rotp          Stellar Rotational Period [days]
209        210          st_radv             Systemic Radial Velocity [km/s]
215        216               ra                                   RA [deg]
217        218              dec                                  Dec [deg]
233        234          sy_dist                              Distance [pc]
```

In [4]:
```
#redoing the internal index column, 0 to 37
new_column_data=new_column_data.reset_index(drop=True)
print(new_column_data)
```

```
    column_id      abbrev_name                               full_name
0           2          pl_name                             Planet Name
1           3         hostname                               Host Name
2           4        pl_letter                           Planet Letter
3          10          sy_snum                         Number of Stars
4          11          sy_pnum                       Number of Planets
5          12          sy_mnum                         Number of Moons
6          14  discoverymethod                        Discovery Method
7          15        disc_year                          Discovery Year
8          19    disc_facility                      Discovery Facility
9          33        pl_orbper                   Orbital Period [days]
10         38       pl_orbsmax               Orbit Semi-Major Axis [au]
11         43        pl_angsep               Angular Separation [mas]
12         48          pl_rade            Planet Radius [Earth Radius]
13         58        pl_bmasse   Planet Mass or Mass*sin(i) [Earth Mass]
14         69          pl_dens               Planet Density [g/cm**3]
15         74       pl_orbeccen                            Eccentricity
16         79         pl_insol             Insolation Flux [Earth Flux]
17         84           pl_eqt             Equilibrium Temperature [K]
18         89       pl_orbincl                       Inclination [deg]
19         94       pl_tranmid                  Transit Midpoint [days]
20        106       pl_trandep                       Transit Depth [%]
21        111       pl_trandur                 Transit Duration [hours]
22        142         pl_rvamp           Radial Velocity Amplitude [m/s]
23        157      st_spectype                           Spectral Type
24        164           st_rad            Stellar Radius [Solar Radius]
25        169          st_mass                Stellar Mass [Solar mass]
26        174           st_met             Stellar Metallicity [dex]
27        180           st_lum           Stellar Luminosity [log(Solar)]
28        185          st_logg  Stellar Surface Gravity [log10(cm/s**2)]
29        190           st_age                         Stellar Age [Gyr]
30        195          st_dens               Stellar Density [g/cm**3]
31        200          st_vsin         Stellar Rotational Velocity [km/s]
32        205          st_rotp         Stellar Rotational Period [days]
33        210          st_radv           Systemic Radial Velocity [km/s]
34        216               ra                                RA [deg]
35        218              dec                               Dec [deg]
36        234          sy_dist                           Distance [pc]
```

In [5]:
```
#redoing the planet's column id, 1 to length of column, 1 to 38
new_column_data['column_id']=range(1, len(new_column_data) + 1)
print(new_column_data)
```

```
    column_id      abbrev_name                                  full_name
0           1          pl_name                                Planet Name
1           2         hostname                                  Host Name
2           3        pl_letter                              Planet Letter
3           4          sy_snum                            Number of Stars
4           5          sy_pnum                          Number of Planets
5           6          sy_mnum                            Number of Moons
6           7  discoverymethod                           Discovery Method
7           8        disc_year                             Discovery Year
8           9    disc_facility                         Discovery Facility
9          10        pl_orbper                      Orbital Period [days]
10         11       pl_orbsmax                 Orbit Semi-Major Axis [au]
11         12        pl_angsep                 Angular Separation [mas]
12         13          pl_rade               Planet Radius [Earth Radius]
13         14        pl_bmasse   Planet Mass or Mass*sin(i) [Earth Mass]
14         15          pl_dens                 Planet Density [g/cm**3]
15         16       pl_orbeccen                              Eccentricity
16         17         pl_insol               Insolation Flux [Earth Flux]
17         18           pl_eqt               Equilibrium Temperature [K]
18         19       pl_orbincl                          Inclination [deg]
19         20       pl_tranmid                   Transit Midpoint [days]
20         21       pl_trandep                          Transit Depth [%]
21         22       pl_trandur                   Transit Duration [hours]
22         23         pl_rvamp            Radial Velocity Amplitude [m/s]
23         24      st_spectype                              Spectral Type
24         25           st_rad               Stellar Radius [Solar Radius]
25         26          st_mass                   Stellar Mass [Solar mass]
26         27           st_met                 Stellar Metallicity [dex]
27         28           st_lum            Stellar Luminosity [log(Solar)]
28         29          st_logg  Stellar Surface Gravity [log10(cm/s**2)]
29         30           st_age                          Stellar Age [Gyr]
30         31          st_dens                Stellar Density [g/cm**3]
31         32          st_vsin         Stellar Rotational Velocity [km/s]
32         33          st_rotp         Stellar Rotational Period [days]
33         34          st_radv           Systemic Radial Velocity [km/s]
34         35               ra                                   RA [deg]
35         36              dec                                  Dec [deg]
36         37          sy_dist                              Distance [pc]
```

In [6]:
```python
#data types and null values for column data
new_column_data.info()
```

```
<class 'pandas.DataFrame'>
RangeIndex: 37 entries, 0 to 36
Data columns (total 3 columns):
 #   Column       Non-Null Count  Dtype
---  ------       --------------  -----
 0   column_id    37 non-null     int64
 1   abbrev_name  37 non-null     str
 2   full_name    37 non-null     str
dtypes: int64(1), str(2)
memory usage: 1020.0 bytes
```

In [7]:
```python
#check for missing values or Na in each column,
#True for missing, false for NOT missing
row_missing = new_column_data.isna()
```

```
#adds the number of rows in each column with missing values
row_total=row_missing.sum()
print(row_total)
```

```
column_id      0
abbrev_name    0
full_name      0
dtype: int64
```

In [8]:
```
#store cleaned data in new CSV file
cleaned_column_data = new_column_data.copy()
cleaned_column_data.to_csv('cleaned_column_data.csv', index=False)
print(cleaned_column_data)
```

```
    column_id      abbrev_name                                full_name
0           1          pl_name                              Planet Name
1           2         hostname                                Host Name
2           3        pl_letter                            Planet Letter
3           4          sy_snum                          Number of Stars
4           5          sy_pnum                        Number of Planets
5           6          sy_mnum                          Number of Moons
6           7  discoverymethod                         Discovery Method
7           8        disc_year                           Discovery Year
8           9    disc_facility                       Discovery Facility
9          10       pl_orbper                     Orbital Period [days]
10         11      pl_orbsmax                 Orbit Semi-Major Axis [au]
11         12       pl_angsep                  Angular Separation [mas]
12         13          pl_rade              Planet Radius [Earth Radius]
13         14        pl_bmasse   Planet Mass or Mass*sin(i) [Earth Mass]
14         15          pl_dens                  Planet Density [g/cm**3]
15         16      pl_orbeccen                             Eccentricity
16         17         pl_insol              Insolation Flux [Earth Flux]
17         18           pl_eqt             Equilibrium Temperature [K]
18         19      pl_orbincl                        Inclination [deg]
19         20      pl_tranmid                  Transit Midpoint [days]
20         21      pl_trandep                        Transit Depth [%]
21         22      pl_trandur                 Transit Duration [hours]
22         23         pl_rvamp             Radial Velocity Amplitude [m/s]
23         24      st_spectype                            Spectral Type
24         25           st_rad              Stellar Radius [Solar Radius]
25         26          st_mass                Stellar Mass [Solar mass]
26         27           st_met                Stellar Metallicity [dex]
27         28           st_lum             Stellar Luminosity [log(Solar)]
28         29          st_logg  Stellar Surface Gravity [log10(cm/s**2)]
29         30           st_age                          Stellar Age [Gyr]
30         31          st_dens                 Stellar Density [g/cm**3]
31         32          st_vsin         Stellar Rotational Velocity [km/s]
32         33          st_rotp         Stellar Rotational Period [days]
33         34          st_radv          Systemic Radial Velocity [km/s]
34         35               ra                                 RA [deg]
35         36              dec                                Dec [deg]
36         37          sy_dist                            Distance [pc]
```

Planet Data

In [9]:
```
#load full planet data file
planet_data=pd.read_csv(r"C:\Users\maybe\OneDrive\Desktop\Databases and Advanced Te


#check first rows to verify it loaded correctly
print(planet_data.head(1))
```

```
   planet_id    pl_name hostname pl_letter    hd_name   hip_name        tic_id  \
0          1  11 Com b   11 Com         b  HD 107383  HIP 60202  TIC 72437047   

                 gaia_dr2_id                   gaia_dr3_id  sy_snum  ...  \
0  Gaia DR2 3946945413106333696  Gaia DR3 3946945413106333696        2  ...   

   sy_kepmagerr1  sy_kepmagerr2  \
0            NaN            NaN   

                                    sy_kepmag_reflink pl_nnotes  st_nphot  \
0  <a refstr=STASSUN_ET_AL__2019 href=https://ui....         2         1   

  st_nrvc st_nspec pl_nespec pl_ntranspec pl_ndispec
0       2        0         0            0          0

[1 rows x 320 columns]
```

```
C:\Users\maybe\AppData\Local\Temp\ipykernel_16316\2617141790.py:2: DtypeWarning: Col
umns (0: pl_imppar_reflink, 1: pl_trandep_reflink, 2: pl_trandur_reflink, 3: pl_ratr
or_reflink, 4: pl_occdep_reflink, 5: pl_projobliq_reflink, 6: pl_trueobliq_reflink)
have mixed types. Specify dtype option on import or set low_memory=False.
  planet_data=pd.read_csv(r"C:\Users\maybe\OneDrive\Desktop\Databases and Advanced T
echniques\full_planets_data.csv")
```

In [10]:
```
### data types and null values for planet data
planet_data.info()
```

```
<class 'pandas.DataFrame'>
RangeIndex: 6158 entries, 0 to 6157
Columns: 320 entries, planet_id to pl_ndispec
dtypes: float64(216), int64(24), str(80)
memory usage: 15.0 MB
```

In [11]:
```
#### Find critical columns
critical_planet_columns=['pl_name', 'hostname', 'pl_letter', 'sy_snum', 'sy_pnum',
                         'sy_mnum','discoverymethod', 'disc_year', 'disc_facility',
                         'pl_orbper','pl_orbsmax', 'pl_angsep' , 'pl_rade', 'pl_bma
                        'pl_dens', 'pl_orbeccen', 'pl_insol', 'pl_eqt', 'pl_orbincl
                        'pl_trandep', 'pl_trandur', 'pl_rvamp', 'st_spectype', 'st_
                        'st_met', 'st_lum', 'st_logg', 'st_age', 'st_dens', 'st_vsi
                        'st_radv', 'ra', 'dec', 'sy_dist']

new_planet_data = planet_data[critical_planet_columns]
print(new_planet_data)
```

```
          pl_name  hostname pl_letter  sy_snum  sy_pnum  sy_mnum  \
0        11 Com b    11 Com         b        2        1        0   
1        11 UMi b    11 UMi         b        1        1        0   
2        14 And b    14 And         b        1        1        0   
3        14 Her b    14 Her         b        1        2        0   
4      16 Cyg B b  16 Cyg B         b        3        1        0   
...           ...       ...       ...      ...      ...      ...   
6153    ups And b   ups And         b        2        3        0   
6154    ups And c   ups And         c        2        3        0   
6155    ups And d   ups And         d        2        3        0   
6156    ups Leo b   ups Leo         b        1        1        0   
6157     xi Aql b    xi Aql         b        1        1        0   

      discoverymethod  disc_year                         disc_facility  \
0     Radial Velocity     2007.0                      Xinglong Station   
1     Radial Velocity     2009.0  Thueringer Landessternwarte Tautenburg   
2     Radial Velocity     2008.0       Okayama Astrophysical Observatory   
3     Radial Velocity     2002.0                  W. M. Keck Observatory   
4     Radial Velocity     1996.0                Multiple Observatories   
...               ...        ...                                   ...   
6153  Radial Velocity     1996.0                      Lick Observatory   
6154  Radial Velocity     1999.0                Multiple Observatories   
6155  Radial Velocity     1999.0                Multiple Observatories   
6156  Radial Velocity     2021.0       Okayama Astrophysical Observatory   
6157  Radial Velocity     2007.0       Okayama Astrophysical Observatory   

         pl_orbper  ...   st_lum   st_logg  st_age   st_dens  st_vsin  st_rotp  \
0       323.210000  ...  1.97823  2.450000     NaN       NaN     1.20      NaN   
1       516.219970  ...  2.42951  1.930000    1.56       NaN     1.50      NaN   
2       186.760000  ...  1.83992  2.550000    4.50       NaN     2.60      NaN   
3      1766.410000  ... -0.15273  4.450000    4.60  1.273928     1.00      NaN   
4       798.500000  ...  0.09729  4.360000    7.40  1.011029     2.70      NaN   
...            ...  ...      ...       ...     ...       ...      ...      ...   
6153      4.617033  ...  0.52541  4.125176    5.00  0.369930     9.20     12.0   
6154    241.258000  ...  0.52541  4.125176    5.00  0.369930     9.20     12.0   
6155   1276.460000  ...  0.52541  4.125176    5.00  0.369930     9.20     12.0   
6156    385.200000  ...  1.80003  2.460000     NaN       NaN     1.78      NaN   
6157    136.970000  ...  1.76745  2.610000    7.10       NaN      NaN      NaN   

       st_radv          ra        dec   sy_dist  
0     43.36898  185.178779  17.793252   93.1846  
1    -17.52023  229.274595  71.823943  125.3210  
2    -59.72633  352.824150  39.235837   75.4392  
3    -13.82260  242.602101  43.816362   17.9323  
4    -28.10000  295.465642  50.516824   21.1397  
...        ...         ...        ...       ...  
6153 -28.65500   24.198353  41.403815   13.4054  
6154 -28.65500   24.198353  41.403815   13.4054  
6155 -28.65500   24.198353  41.403815   13.4054  
6156       NaN  174.237219  -0.823564   52.5973  
6157 -41.66164  298.562449   8.461105   56.1858  

[6158 rows x 37 columns]
```

In [12]:

```
#check for missing values or Na in each column,
#True for missing, false for NOT missing
```

```
planet_missing = new_planet_data.isna()

#adds the number of rows in each column with missing values
row_total=planet_missing.sum()
print(row_total)
```

```
pl_name              0
hostname             0
pl_letter            0
sy_snum              0
sy_pnum              0
sy_mnum              0
discoverymethod      0
disc_year            1
disc_facility        0
pl_orbper          337
pl_orbsmax         315
pl_angsep          344
pl_rade             50
pl_bmasse           31
pl_dens            139
pl_orbeccen        938
pl_insol          1842
pl_eqt            1565
pl_orbincl        1522
pl_tranmid        1266
pl_trandep        1777
pl_trandur        1683
pl_rvamp          3667
st_spectype       3847
st_rad             317
st_mass              8
st_met             553
st_lum             311
st_logg            321
st_age            1314
st_dens            586
st_vsin           3881
st_rotp           5268
st_radv           3772
ra                   0
dec                  0
sy_dist             27
dtype: int64
```

In [13]:

```
#we need to drop all planets with missing values under a column, starting with pl_o
required_columns = ['disc_year','pl_orbper','pl_orbsmax', 'pl_angsep' , 'pl_rade',
                        'pl_dens', 'pl_orbeccen', 'pl_insol', 'pl_eqt', 'pl_orbincl
                        'pl_trandep', 'pl_trandur', 'pl_rvamp', 'st_spectype', 'st_
                        'st_met', 'st_lum', 'st_logg', 'st_age', 'st_dens', 'st_vsi
                        'st_radv', 'ra', 'dec', 'sy_dist']

new_planet_data.dropna(subset= required_columns, inplace=True)

#calculating columns with missing values again after dropping planets missing data
#all are zeros
```

```
missing_row_data=new_planet_data.isna()
total_rows=missing_row_data.sum()
print(total_rows)
```

```
pl_name            0
hostname           0
pl_letter          0
sy_snum            0
sy_pnum            0
sy_mnum            0
discoverymethod    0
disc_year          0
disc_facility      0
pl_orbper          0
pl_orbsmax         0
pl_angsep          0
pl_rade            0
pl_bmasse          0
pl_dens            0
pl_orbeccen        0
pl_insol           0
pl_eqt             0
pl_orbincl         0
pl_tranmid         0
pl_trandep         0
pl_trandur         0
pl_rvamp           0
st_spectype        0
st_rad             0
st_mass            0
st_met             0
st_lum             0
st_logg            0
st_age             0
st_dens            0
st_vsin            0
st_rotp            0
st_radv            0
ra                 0
dec                0
sy_dist            0
dtype: int64
```

In [14]:

```
#calculating rows with missing values again after dropping planets missing data
#all are zeros
missing_column_data = new_planet_data.isna()
total_columns = missing_column_data.sum(axis=1)
#left with 162 entries, or 162 planets that have data in all the required columns
print(total_columns)
```

```
36      0
54      0
55      0
110     0
209     0
       ..
6078    0
6097    0
6100    0
6107    0
6141    0
Length: 162, dtype: int64
```

In [15]:
```
#check to see if rows are full
print(new_planet_data)
```

```
          pl_name  hostname pl_letter  sy_snum  sy_pnum  sy_mnum  \
36       55 Cnc e    55 Cnc         e        2        7        0
54       AU Mic b    AU Mic         b        1        4        0
55       AU Mic c    AU Mic         c        1        4        0
110    CoRoT-18 b  CoRoT-18         b        1        1        0
209      G 9-40 b    G 9-40         b        1        1        0
...           ...       ...       ...      ...      ...      ...
6078    WASP-96 b   WASP-96         b        1        1        0
6097   Wolf 327 b  Wolf 327         b        1        1        0
6100     XO-2 N b    XO-2 N         b        2        3        0
6107       XO-7 b      XO-7         b        1        1        0
6141     pi Men c  HD 39091         c        1        3        0

      discoverymethod  disc_year  \
36    Radial Velocity     2004.0
54            Transit     2020.0
55            Transit     2021.0
110           Transit     2011.0
209           Transit     2019.0
...               ...        ...
6078          Transit     2014.0
6097          Transit     2024.0
6100          Transit     2007.0
6107          Transit     2019.0
6141          Transit     2018.0

                                  disc_facility  pl_orbper  ...   st_lum  \
36                        McDonald Observatory   0.736547  ... -0.19695
54    Transiting Exoplanet Survey Satellite (TESS)   8.463446  ... -0.97757
55    Transiting Exoplanet Survey Satellite (TESS)  18.859023  ... -0.97757
110                                       CoRoT   1.900069  ... -0.17000
209                                          K2   5.745998  ... -1.96019
...                                         ...        ...  ...      ...
6078                                  SuperWASP   3.425260  ...  0.03226
6097  Transiting Exoplanet Survey Satellite (TESS)   0.573474  ... -1.62986
6100                                         XO   2.615862  ... -0.08823
6107                                         XO   2.864142  ...  0.43132
6141  Transiting Exoplanet Survey Satellite (TESS)   6.267840  ...  0.15836

      st_logg  st_age   st_dens  st_vsin  st_rotp   st_radv          ra  \
36      4.430   5.500  1.521203     1.23    38.80  27.40949  133.146837
54      4.370   0.019  1.750000     9.20     4.86  -6.31031  311.291137
55      4.370   0.019  1.750000     9.20     4.86  -6.31031  311.291137
110     4.400   0.600  1.350000     8.00     5.40  29.55000   98.172417
209     4.840   9.900  1.620000     2.00    30.00  14.65000  134.718851
...       ...     ...       ...      ...      ...       ...         ...
6078    4.420   8.000  1.299000     1.50    20.16  -1.23300    1.046570
6097    4.870   4.100  8.540000     2.00    44.40   9.39000  148.378320
6100    4.433   7.800  1.410000     1.07    28.60  46.93290  117.026769
6107    4.246   1.180  0.451084     6.00     3.62 -12.98300  277.478075
6141    4.430   2.980  0.944584     3.16    18.30  10.70916   84.299280

            dec   sy_dist
36    28.329815   12.5855
54   -31.342450    9.7221
55   -31.342450    9.7221
```

```
110   -0.031605  764.8890
209   21.074796   27.9276
...         ...       ...
6078 -47.360635  352.4640
6097  35.570548   28.5321
6100  50.225147  154.2730
6107  85.233321  234.1490
6141 -80.464604   18.2702

[162 rows x 37 columns]
```

In [16]:

```
#redoing the internal index column, 0 to 70
new_planet_data=new_planet_data.reset_index(drop=True)
print(new_planet_data)
```

```
         pl_name  hostname pl_letter  sy_snum  sy_pnum  sy_mnum  \
0      55 Cnc e    55 Cnc         e        2        7        0
1      AU Mic b    AU Mic         b        1        4        0
2      AU Mic c    AU Mic         c        1        4        0
3    CoRoT-18 b  CoRoT-18         b        1        1        0
4      G 9-40 b    G 9-40         b        1        1        0
..          ...       ...       ...      ...      ...      ...
157   WASP-96 b   WASP-96         b        1        1        0
158  Wolf 327 b  Wolf 327         b        1        1        0
159    XO-2 N b    XO-2 N         b        2        3        0
160      XO-7 b      XO-7         b        1        1        0
161    pi Men c  HD 39091         c        1        3        0

    discoverymethod  disc_year                                 disc_facility  \
0   Radial Velocity     2004.0                         McDonald Observatory
1           Transit     2020.0  Transiting Exoplanet Survey Satellite (TESS)
2           Transit     2021.0  Transiting Exoplanet Survey Satellite (TESS)
3           Transit     2011.0                                         CoRoT
4           Transit     2019.0                                            K2
..              ...        ...                                           ...
157         Transit     2014.0                                     SuperWASP
158         Transit     2024.0  Transiting Exoplanet Survey Satellite (TESS)
159         Transit     2007.0                                            XO
160         Transit     2019.0                                            XO
161         Transit     2018.0  Transiting Exoplanet Survey Satellite (TESS)

     pl_orbper  ...   st_lum  st_logg  st_age   st_dens  st_vsin  st_rotp  \
0     0.736547  ... -0.19695    4.430   5.500  1.521203     1.23    38.80
1     8.463446  ... -0.97757    4.370   0.019  1.750000     9.20     4.86
2    18.859023  ... -0.97757    4.370   0.019  1.750000     9.20     4.86
3     1.900069  ... -0.17000    4.400   0.600  1.350000     8.00     5.40
4     5.745998  ... -1.96019    4.840   9.900  1.620000     2.00    30.00
..         ...  ...      ...      ...     ...       ...      ...      ...
157   3.425260  ...  0.03226    4.420   8.000  1.299000     1.50    20.16
158   0.573474  ... -1.62986    4.870   4.100  8.540000     2.00    44.40
159   2.615862  ... -0.08823    4.433   7.800  1.410000     1.07    28.60
160   2.864142  ...  0.43132    4.246   1.180  0.451084     6.00     3.62
161   6.267840  ...  0.15836    4.430   2.980  0.944584     3.16    18.30

      st_radv          ra        dec   sy_dist
0    27.40949  133.146837  28.329815   12.5855
1    -6.31031  311.291137 -31.342450    9.7221
2    -6.31031  311.291137 -31.342450    9.7221
3    29.55000   98.172417  -0.031605  764.8890
4    14.65000  134.718851  21.074796   27.9276
..        ...         ...        ...       ...
157  -1.23300    1.046570 -47.360635  352.4640
158   9.39000  148.378320  35.570548   28.5321
159  46.93290  117.026769  50.225147  154.2730
160 -12.98300  277.478075  85.233321  234.1490
161  10.70916   84.299280 -80.464604   18.2702

[162 rows x 37 columns]
```

In [17]:

```
#renaming the planet's id with new table, 1 to length of column, 1 to 71
#new_planet_data['planet_id']=range(1, len(new_planet_data) + 1)
```

```
print(new_planet_data)
```

```
        pl_name   hostname pl_letter  sy_snum  sy_pnum  sy_mnum  \
0      55 Cnc e     55 Cnc         e        2        7        0   
1      AU Mic b     AU Mic         b        1        4        0   
2      AU Mic c     AU Mic         c        1        4        0   
3    CoRoT-18 b   CoRoT-18         b        1        1        0   
4      G 9-40 b     G 9-40         b        1        1        0   
..          ...        ...       ...      ...      ...      ...   
157   WASP-96 b    WASP-96         b        1        1        0   
158  Wolf 327 b   Wolf 327         b        1        1        0   
159    XO-2 N b     XO-2 N         b        2        3        0   
160      XO-7 b       XO-7         b        1        1        0   
161    pi Men c   HD 39091         c        1        3        0   

    discoverymethod  disc_year                                disc_facility  \
0   Radial Velocity     2004.0                          McDonald Observatory   
1           Transit     2020.0  Transiting Exoplanet Survey Satellite (TESS)   
2           Transit     2021.0  Transiting Exoplanet Survey Satellite (TESS)   
3           Transit     2011.0                                         CoRoT   
4           Transit     2019.0                                            K2   
..              ...        ...                                           ...   
157         Transit     2014.0                                     SuperWASP   
158         Transit     2024.0  Transiting Exoplanet Survey Satellite (TESS)   
159         Transit     2007.0                                            XO   
160         Transit     2019.0                                            XO   
161         Transit     2018.0  Transiting Exoplanet Survey Satellite (TESS)   

      pl_orbper  ...   st_lum  st_logg  st_age   st_dens  st_vsin  st_rotp  \
0      0.736547  ... -0.19695    4.430   5.500  1.521203     1.23    38.80   
1      8.463446  ... -0.97757    4.370   0.019  1.750000     9.20     4.86   
2     18.859023  ... -0.97757    4.370   0.019  1.750000     9.20     4.86   
3      1.900069  ... -0.17000    4.400   0.600  1.350000     8.00     5.40   
4      5.745998  ... -1.96019    4.840   9.900  1.620000     2.00    30.00   
..          ...  ...      ...      ...     ...       ...      ...      ...   
157    3.425260  ...  0.03226    4.420   8.000  1.299000     1.50    20.16   
158    0.573474  ... -1.62986    4.870   4.100  8.540000     2.00    44.40   
159    2.615862  ... -0.08823    4.433   7.800  1.410000     1.07    28.60   
160    2.864142  ...  0.43132    4.246   1.180  0.451084     6.00     3.62   
161    6.267840  ...  0.15836    4.430   2.980  0.944584     3.16    18.30   

      st_radv          ra        dec   sy_dist  
0    27.40949  133.146837  28.329815   12.5855  
1    -6.31031  311.291137 -31.342450    9.7221  
2    -6.31031  311.291137 -31.342450    9.7221  
3    29.55000   98.172417  -0.031605  764.8890  
4    14.65000  134.718851  21.074796   27.9276  
..        ...         ...        ...       ...  
157  -1.23300    1.046570 -47.360635  352.4640  
158   9.39000  148.378320  35.570548   28.5321  
159  46.93290  117.026769  50.225147  154.2730  
160 -12.98300  277.478075  85.233321  234.1490  
161  10.70916   84.299280 -80.464604   18.2702  

[162 rows x 37 columns]
```

Stage 2. Model your data

In [18]:

```
## instead of having two datasets, make the columns names into their full names for

#rename columns here
new_planet_data = new_planet_data.rename(columns={'pl_name': 'Planet Name',
                                                  'hostname': 'Host Name', 'pl_lett
                                                  'sy_snum':'Number of Stars', 'sy_
                                                  'sy_mnum':'Number of Moons','disc
                                                  'disc_year':'Discovery Year','dis
                                                  'pl_orbper':'Orbital Period [days
                                                  'pl_angsep':'Angular Separation [
                                                  'pl_bmasse':'Planet Mass or Mass*
                                                  'pl_orbeccen': 'Eccentricity', 'p
                                                  'pl_eqt': 'Equilibrium Temperatur
                                                  'pl_tranmid':'Transit Midpoint [d
                                                  'pl_trandur':'Transit Duration [h
                                                  'st_spectype':'Spectral Type', 's
                                                  'st_mass':'Stellar Mass [Solar ma
                                                  'st_lum':'Stellar Luminosity [log
                                                  'st_age':'Stellar Age [Gyr]', 'st
                                                  'st_vsin':'Stellar Rotational Vel
                                                  'st_radv':'Systemic Radial Veloci
                                                  'sy_dist':'Distance [pc]'})
print(new_planet_data)
```

```
    Planet Name Host Name Planet Letter  Number of Stars  Number of Planets  \
0      55 Cnc e    55 Cnc             e                2                  7   
1      AU Mic b    AU Mic             b                1                  4   
2      AU Mic c    AU Mic             c                1                  4   
3    CoRoT-18 b  CoRoT-18             b                1                  1   
4      G 9-40 b    G 9-40             b                1                  1   
..          ...       ...           ...              ...                ...   
157   WASP-96 b   WASP-96             b                1                  1   
158  Wolf 327 b  Wolf 327             b                1                  1   
159    XO-2 N b    XO-2 N             b                2                  3   
160      XO-7 b      XO-7             b                1                  1   
161    pi Men c  HD 39091             c                1                  3   

     Number of Moons Discovery Method  Discovery Year  \
0                  0  Radial Velocity          2004.0   
1                  0          Transit          2020.0   
2                  0          Transit          2021.0   
3                  0          Transit          2011.0   
4                  0          Transit          2019.0   
..               ...              ...             ...   
157                0          Transit          2014.0   
158                0          Transit          2024.0   
159                0          Transit          2007.0   
160                0          Transit          2019.0   
161                0          Transit          2018.0   

                               Discovery Facility  Orbital Period [days]  ...  \
0                            McDonald Observatory               0.736547  ...   
1    Transiting Exoplanet Survey Satellite (TESS)               8.463446  ...   
2    Transiting Exoplanet Survey Satellite (TESS)              18.859023  ...   
3                                           CoRoT               1.900069  ...   
4                                              K2               5.745998  ...   
..                                            ...                    ...  ...   
157                                     SuperWASP               3.425260  ...   
158  Transiting Exoplanet Survey Satellite (TESS)               0.573474  ...   
159                                            XO               2.615862  ...   
160                                            XO               2.864142  ...   
161  Transiting Exoplanet Survey Satellite (TESS)               6.267840  ...   

     Stellar Luminosity [log(Solar)]  \
0                           -0.19695   
1                           -0.97757   
2                           -0.97757   
3                           -0.17000   
4                           -1.96019   
..                               ...   
157                          0.03226   
158                         -1.62986   
159                         -0.08823   
160                          0.43132   
161                          0.15836   

     Stellar Surface Gravity [log10(cm/s**2)]  Stellar Age [Gyr]  \
0                                       4.430              5.500   
1                                       4.370              0.019   
2                                       4.370              0.019   
```

```
3                                        4.400              0.600
4                                        4.840              9.900
..                                         ...                ...
157                                      4.420              8.000
158                                      4.870              4.100
159                                      4.433              7.800
160                                      4.246              1.180
161                                      4.430              2.980

     Stellar Density [g/cm**3]  Stellar Rotational Velocity [km/s]  \
0                     1.521203                                1.23
1                     1.750000                                9.20
2                     1.750000                                9.20
3                     1.350000                                8.00
4                     1.620000                                2.00
..                         ...                                 ...
157                   1.299000                                1.50
158                   8.540000                                2.00
159                   1.410000                                1.07
160                   0.451084                                6.00
161                   0.944584                                3.16

     Stellar Rotational Period [days]  Systemic Radial Velocity [km/s]  \
0                               38.80                         27.40949
1                                4.86                         -6.31031
2                                4.86                         -6.31031
3                                5.40                         29.55000
4                               30.00                         14.65000
..                                ...                              ...
157                             20.16                         -1.23300
158                             44.40                          9.39000
159                             28.60                         46.93290
160                              3.62                        -12.98300
161                             18.30                         10.70916

       RA [deg]  Dec [deg]  Distance [pc]
0    133.146837  28.329815        12.5855
1    311.291137 -31.342450         9.7221
2    311.291137 -31.342450         9.7221
3     98.172417  -0.031605       764.8890
4    134.718851  21.074796        27.9276
..          ...        ...            ...
157    1.046570 -47.360635       352.4640
158  148.378320  35.570548        28.5321
159  117.026769  50.225147       154.2730
160  277.478075  85.233321       234.1490
161   84.299280 -80.464604        18.2702

[162 rows x 37 columns]
```

In [19]:

```
## ************** foreign keys columns do not have 1 to n in sequential order *****

##add new columns to dataset, Star ID, Discovery ID, Solar System ID

#variables for number of columns and rows or current data
```

```
#number_columns = len(new_planet_data.columns)
#number_rows = len(new_planet_data)

## add new columns using data.insert(location, column name, rows) at the end of the

#new_planet_data.insert(number_columns, 'Star ID', range(1,  number_rows + 1))

#new_planet_data.insert(number_columns + 1, 'System ID', range(1, number_rows + 1))

#new_planet_data.insert(number_columns + 2, 'Discovery ID', range(1, number_rows +


print(new_planet_data)
```

```
    Planet Name Host Name Planet Letter  Number of Stars  Number of Planets  \
0      55 Cnc e    55 Cnc             e                2                  7   
1      AU Mic b    AU Mic             b                1                  4   
2      AU Mic c    AU Mic             c                1                  4   
3    CoRoT-18 b  CoRoT-18             b                1                  1   
4      G 9-40 b    G 9-40             b                1                  1   
..          ...       ...           ...              ...                ...   
157   WASP-96 b   WASP-96             b                1                  1   
158  Wolf 327 b  Wolf 327             b                1                  1   
159    XO-2 N b    XO-2 N             b                2                  3   
160      XO-7 b      XO-7             b                1                  1   
161    pi Men c  HD 39091             c                1                  3   

     Number of Moons Discovery Method  Discovery Year  \
0                  0  Radial Velocity          2004.0   
1                  0          Transit          2020.0   
2                  0          Transit          2021.0   
3                  0          Transit          2011.0   
4                  0          Transit          2019.0   
..               ...              ...             ...   
157                0          Transit          2014.0   
158                0          Transit          2024.0   
159                0          Transit          2007.0   
160                0          Transit          2019.0   
161                0          Transit          2018.0   

                               Discovery Facility  Orbital Period [days]  ...  \
0                            McDonald Observatory               0.736547  ...   
1    Transiting Exoplanet Survey Satellite (TESS)               8.463446  ...   
2    Transiting Exoplanet Survey Satellite (TESS)              18.859023  ...   
3                                           CoRoT               1.900069  ...   
4                                              K2               5.745998  ...   
..                                            ...                    ...  ...   
157                                     SuperWASP               3.425260  ...   
158  Transiting Exoplanet Survey Satellite (TESS)               0.573474  ...   
159                                            XO               2.615862  ...   
160                                            XO               2.864142  ...   
161  Transiting Exoplanet Survey Satellite (TESS)               6.267840  ...   

     Stellar Luminosity [log(Solar)]  \
0                           -0.19695   
1                           -0.97757   
2                           -0.97757   
3                           -0.17000   
4                           -1.96019   
..                               ...   
157                          0.03226   
158                         -1.62986   
159                         -0.08823   
160                          0.43132   
161                          0.15836   

     Stellar Surface Gravity [log10(cm/s**2)]  Stellar Age [Gyr]  \
0                                       4.430              5.500   
1                                       4.370              0.019   
2                                       4.370              0.019   
```

```
3                                        4.400            0.600
4                                        4.840            9.900
..                                         ...              ...
157                                      4.420            8.000
158                                      4.870            4.100
159                                      4.433            7.800
160                                      4.246            1.180
161                                      4.430            2.980

     Stellar Density [g/cm**3]  Stellar Rotational Velocity [km/s]  \
0                     1.521203                                1.23
1                     1.750000                                9.20
2                     1.750000                                9.20
3                     1.350000                                8.00
4                     1.620000                                2.00
..                         ...                                 ...
157                   1.299000                                1.50
158                   8.540000                                2.00
159                   1.410000                                1.07
160                   0.451084                                6.00
161                   0.944584                                3.16

     Stellar Rotational Period [days]  Systemic Radial Velocity [km/s]  \
0                               38.80                         27.40949
1                                4.86                         -6.31031
2                                4.86                         -6.31031
3                                5.40                         29.55000
4                               30.00                         14.65000
..                                ...                              ...
157                             20.16                         -1.23300
158                             44.40                          9.39000
159                             28.60                         46.93290
160                              3.62                        -12.98300
161                             18.30                         10.70916

       RA [deg]  Dec [deg]  Distance [pc]
0    133.146837  28.329815        12.5855
1    311.291137 -31.342450         9.7221
2    311.291137 -31.342450         9.7221
3     98.172417  -0.031605       764.8890
4    134.718851  21.074796        27.9276
..          ...        ...            ...
157    1.046570 -47.360635       352.4640
158  148.378320  35.570548        28.5321
159  117.026769  50.225147       154.2730
160  277.478075  85.233321       234.1490
161   84.299280 -80.464604        18.2702

[162 rows x 37 columns]
```

In [20]:

```
## Create a new dataset for Stars, Solar System, and Discoveries

#stars_data = new_planet_data['Star ID']
#print(stars_data)
```

```
#system_data =  new_planet_data['System ID']
#print(system_data)

#discoveries_data =  new_planet_data['Discovery ID']
#print(discoveries_data)
```

In [21]:
```
## Add data to new data sets
#i need to find the columns, then i need add the columns to the new dataset

#Stars
#Spectral Type, Stellar Radius [Solar Radius],
#Stellar Mass [Solar mass], Stellar Metallicity [dex],
#Stellar Luminosity [log(Solar)], Stellar Surface Gravity [log10(cm/s**2)],
#Stellar Age [Gyr], Stellar Density [g/cm**3], Stellar Rotational Velocity [km/s]
#Stellar Rotational Period [days], Systemic Radial Velocity [km/s],

#find columns and assign to variable
stars_columns=['Host Name','Spectral Type', 'Stellar Radius [Solar Radius]',
'Stellar Mass [Solar mass]', 'Stellar Metallicity [dex]',
'Stellar Luminosity [log(Solar)]', 'Stellar Surface Gravity [log10(cm/s**2)]',
'Stellar Age [Gyr]', 'Stellar Density [g/cm**3]', 'Stellar Rotational Velocity [km/
'Stellar Rotational Period [days]', 'Systemic Radial Velocity [km/s]']

#redefine stars_data using new varable with column names
stars_data = new_planet_data[stars_columns]

#set variable number of rows in dataset
#number_rows = len(stars_data)

## add new columns using data.insert(location, column name, rows) at the end of the
#stars_data.insert(0, 'Star ID', range(1,  number_rows + 1))


print(stars_data)
```

```
    Host Name Spectral Type  Stellar Radius [Solar Radius]  \
0      55 Cnc          G8 V                         0.9430   
1      AU Mic          M1 V                         0.8620   
2      AU Mic          M1 V                         0.8620   
3    CoRoT-18          G9 V                         1.0000   
4      G 9-40        M2.5 V                         0.3026   
..        ...           ...                            ...   
157   WASP-96            G8                         1.0500   
158  Wolf 327        M2.5 V                         0.4060   
159    XO-2 N          K0 V                         0.9900   
160      XO-7          G0 V                         1.4800   
161  HD 39091          G0 V                         1.1700   

     Stellar Mass [Solar mass]  Stellar Metallicity [dex]  \
0                       0.9050                      0.350   
1                       0.6350                      0.010   
2                       0.6350                      0.010   
3                       0.9500                     -0.100   
4                       0.2952                     -0.070   
..                         ...                        ...   
157                     1.0600                      0.140   
158                     0.4050                     -0.170   
159                     0.9710                      0.430   
160                     1.4050                      0.432   
161                     1.0700                      0.090   

     Stellar Luminosity [log(Solar)]  \
0                           -0.19695   
1                           -0.97757   
2                           -0.97757   
3                           -0.17000   
4                           -1.96019   
..                               ...   
157                          0.03226   
158                         -1.62986   
159                         -0.08823   
160                          0.43132   
161                          0.15836   

     Stellar Surface Gravity [log10(cm/s**2)]  Stellar Age [Gyr]  \
0                                       4.430              5.500   
1                                       4.370              0.019   
2                                       4.370              0.019   
3                                       4.400              0.600   
4                                       4.840              9.900   
..                                        ...                ...   
157                                     4.420              8.000   
158                                     4.870              4.100   
159                                     4.433              7.800   
160                                     4.246              1.180   
161                                     4.430              2.980   

     Stellar Density [g/cm**3]  Stellar Rotational Velocity [km/s]  \
0                     1.521203                                1.23   
1                     1.750000                                9.20   
2                     1.750000                                9.20   
```

```
3                    1.350000                                     8.00
4                    1.620000                                     2.00
..                        ...                                      ...
157                  1.299000                                     1.50
158                  8.540000                                     2.00
159                  1.410000                                     1.07
160                  0.451084                                     6.00
161                  0.944584                                     3.16

     Stellar Rotational Period [days]  Systemic Radial Velocity [km/s]
0                               38.80                         27.40949
1                                4.86                         -6.31031
2                                4.86                         -6.31031
3                                5.40                         29.55000
4                               30.00                         14.65000
..                                ...                              ...
157                             20.16                         -1.23300
158                             44.40                          9.39000
159                             28.60                         46.93290
160                              3.62                        -12.98300
161                             18.30                         10.70916

[162 rows x 12 columns]
```

In [22]:

```python
## there is a foreign key in stars data that links it to solar system
# we need to add system_id at the end of the dataset like we need for exoplanets ab


number_rows = len(stars_data)

## add new columns using data.insert(location, column name, rows)
stars_data.insert(1, 'System ID', range(1, number_rows + 1))

print(stars_data)
```

```
    Host Name  System ID Spectral Type  Stellar Radius [Solar Radius]  \
0      55 Cnc          1          G8 V                         0.9430   
1      AU Mic          2          M1 V                         0.8620   
2      AU Mic          3          M1 V                         0.8620   
3    CoRoT-18          4          G9 V                         1.0000   
4      G 9-40          5        M2.5 V                         0.3026   
..        ...        ...           ...                            ...   
157   WASP-96        158            G8                         1.0500   
158  Wolf 327        159        M2.5 V                         0.4060   
159    XO-2 N        160          K0 V                         0.9900   
160      XO-7        161          G0 V                         1.4800   
161  HD 39091        162          G0 V                         1.1700   

     Stellar Mass [Solar mass]  Stellar Metallicity [dex]  \
0                       0.9050                      0.350   
1                       0.6350                      0.010   
2                       0.6350                      0.010   
3                       0.9500                     -0.100   
4                       0.2952                     -0.070   
..                         ...                        ...   
157                     1.0600                      0.140   
158                     0.4050                     -0.170   
159                     0.9710                      0.430   
160                     1.4050                      0.432   
161                     1.0700                      0.090   

     Stellar Luminosity [log(Solar)]  \
0                           -0.19695   
1                           -0.97757   
2                           -0.97757   
3                           -0.17000   
4                           -1.96019   
..                               ...   
157                          0.03226   
158                         -1.62986   
159                         -0.08823   
160                          0.43132   
161                          0.15836   

     Stellar Surface Gravity [log10(cm/s**2)]  Stellar Age [Gyr]  \
0                                       4.430              5.500   
1                                       4.370              0.019   
2                                       4.370              0.019   
3                                       4.400              0.600   
4                                       4.840              9.900   
..                                        ...                ...   
157                                     4.420              8.000   
158                                     4.870              4.100   
159                                     4.433              7.800   
160                                     4.246              1.180   
161                                     4.430              2.980   

     Stellar Density [g/cm**3]  Stellar Rotational Velocity [km/s]  \
0                     1.521203                                1.23   
1                     1.750000                                9.20   
2                     1.750000                                9.20   
```

```
3                    1.350000                                   8.00
4                    1.620000                                   2.00
..                        ...                                    ...
157                  1.299000                                   1.50
158                  8.540000                                   2.00
159                  1.410000                                   1.07
160                  0.451084                                   6.00
161                  0.944584                                   3.16

     Stellar Rotational Period [days]  Systemic Radial Velocity [km/s]
0                               38.80                         27.40949
1                                4.86                         -6.31031
2                                4.86                         -6.31031
3                                5.40                         29.55000
4                               30.00                         14.65000
..                                ...                              ...
157                             20.16                         -1.23300
158                             44.40                          9.39000
159                             28.60                         46.93290
160                              3.62                        -12.98300
161                             18.30                         10.70916

[162 rows x 13 columns]
```

In [23]:

```
#System
#System ID, Host Name, Number of Stars, Number of Planets, RA [deg],
#Dec [deg], Distance [pc]

system_columns=[ 'Host Name', 'Number of Stars', 'Number of Planets', 'RA [deg]',
'Dec [deg]', 'Distance [pc]']

system_data = new_planet_data[system_columns]

#set variable number of rows in dataset
#number_rows = len(system_data)

## add new columns using data.insert(location, column name, rows) at the end of the
#system_data.insert(0, 'System ID', range(1,  number_rows + 1))

print(system_data)
```

```
    Host Name  Number of Stars  Number of Planets    RA [deg]  Dec [deg]  \
0      55 Cnc                2                  7  133.146837  28.329815
1      AU Mic                1                  4  311.291137 -31.342450
2      AU Mic                1                  4  311.291137 -31.342450
3    CoRoT-18                1                  1   98.172417  -0.031605
4      G 9-40                1                  1  134.718851  21.074796
..        ...              ...                ...         ...        ...
157   WASP-96                1                  1    1.046570 -47.360635
158  Wolf 327                1                  1  148.378320  35.570548
159    XO-2 N                2                  3  117.026769  50.225147
160      XO-7                1                  1  277.478075  85.233321
161  HD 39091                1                  3   84.299280 -80.464604

     Distance [pc]
0          12.5855
1           9.7221
2           9.7221
3         764.8890
4          27.9276
..             ...
157       352.4640
158        28.5321
159       154.2730
160       234.1490
161        18.2702

[162 rows x 6 columns]
```

In [24]:
```
#Discoveries
#Discovery ID, Discovery Method, Discovery Year, Discovery Facility

discovery_columns=['Host Name','Discovery Method', 'Discovery Year', 'Discovery Fac

discoveries_data = new_planet_data[discovery_columns]

#set variable number of rows in dataset
#number_rows = len(discoveries_data)

## add new columns using data.insert(location, column name, rows) at the end of the
#discoveries_data.insert(0, 'Discovery ID', range(1,  number_rows + 1))

print(discoveries_data)
```

```
    Host Name Discovery Method  Discovery Year  \
0      55 Cnc  Radial Velocity          2004.0
1      AU Mic          Transit          2020.0
2      AU Mic          Transit          2021.0
3    CoRoT-18          Transit          2011.0
4      G 9-40          Transit          2019.0
..        ...              ...             ...
157   WASP-96          Transit          2014.0
158  Wolf 327          Transit          2024.0
159    XO-2 N          Transit          2007.0
160      XO-7          Transit          2019.0
161  HD 39091          Transit          2018.0

                               Discovery Facility
0                           McDonald Observatory
1    Transiting Exoplanet Survey Satellite (TESS)
2    Transiting Exoplanet Survey Satellite (TESS)
3                                           CoRoT
4                                              K2
..                                            ...
157                                     SuperWASP
158  Transiting Exoplanet Survey Satellite (TESS)
159                                            XO
160                                            XO
161  Transiting Exoplanet Survey Satellite (TESS)

[162 rows x 4 columns]
```

In [25]:
```python
#Planet ID, Planet Name, Planet Letter, Number of Moons, Orbital Period [days], Orb
#Angular Separation [mas], Planet Radius [Earth Radius], Planet Mass or Mass*sin(i)
#Eccentricity, Insolation Flux [Earth Flux], Equilibrium Temperature [K], Inclinati
#Transit Depth [%], Transit Duration [hours], Radial Velocity Amplitude [m/s], Star

planet_columns=['Host Name','Planet Name', 'Planet Letter', 'Number of Moons', 'Orb
'Angular Separation [mas]', 'Planet Radius [Earth Radius]', 'Planet Mass or Mass*si
'Eccentricity', 'Insolation Flux [Earth Flux]', 'Equilibrium Temperature [K]', 'Inc
'Transit Depth [%]', 'Transit Duration [hours]', 'Radial Velocity Amplitude [m/s]']

new_planet_data = new_planet_data[planet_columns]
print(new_planet_data)
```

```
    Host Name Planet Name Planet Letter  Number of Moons  \
0      55 Cnc    55 Cnc e             e                0
1      AU Mic    AU Mic b             b                0
2      AU Mic    AU Mic c             c                0
3    CoRoT-18  CoRoT-18 b             b                0
4      G 9-40    G 9-40 b             b                0
..        ...         ...           ...              ...
157   WASP-96   WASP-96 b             b                0
158  Wolf 327  Wolf 327 b             b                0
159    XO-2 N    XO-2 N b             b                0
160      XO-7      XO-7 b             b                0
161  HD 39091    pi Men c             c                0

     Orbital Period [days]  Orbit Semi-Major Axis [au]  \
0                 0.736547                     0.01544
1                 8.463446                     0.07000
2                18.859023                     0.11900
3                 1.900069                     0.02950
4                 5.745998                     0.04180
..                     ...                         ...
157               3.425260                     0.04530
158               0.573474                     0.01000
159               2.615862                     0.03680
160               2.864142                     0.04421
161               6.267840                     0.06839

     Angular Separation [mas]  Planet Radius [Earth Radius]  \
0                      1.2300                      1.875000
1                      7.2000                      4.790000
2                     12.2000                      2.790000
3                      0.0386                     14.680000
4                      1.5000                      1.900000
..                        ...                           ...
157                    0.1290                     13.450000
158                    0.3500                      1.240000
159                    0.2390                     11.130537
160                    0.1890                     15.389957
161                    3.7400                      2.018900

     Planet Mass or Mass*sin(i) [Earth Mass]  Planet Density [g/cm**3]  \
0                                    7.99000                     6.660
1                                    8.99000                     0.490
2                                   14.46000                     3.900
3                                 1102.82000                     2.200
4                                    4.00000                     3.200
..                                       ...                       ...
157                                152.55000                     0.370
158                                  2.53000                     7.240
159                                179.89178                     0.715
160                                225.34147                     0.340
161                                  3.63000                     2.800

     Eccentricity  Insolation Flux [Earth Flux]  Equilibrium Temperature [K]  \
0           0.050                     2657.8322                       1958.0
1           0.070                       21.2000                        554.8
2           0.180                        7.3000                        424.7
```

```
3           0.080                   823.7009                     1550.0
4           0.000                     6.2700                      440.6
..            ...                        ...                        ...
157         0.000                   517.0000                     1285.0
158         0.000                   233.9000                      996.0
159         0.028                   428.0360                     1328.0
160         0.038                  1540.0000                     1743.0
161         0.000                   309.0000                     1169.8

     Inclination [deg]  Transit Midpoint [days]  Transit Depth [%]  \
0               83.590              2457063.210           0.038467
1               88.390              2458330.351           0.249000
2               89.460              2458342.223           0.085000
3               86.500              2455321.724           2.571701
4               89.030              2459503.327           0.365000
..                 ...                      ...                ...
157             85.600              2456258.062           1.380000
158             84.890              2459252.981           0.081839
159             88.010              2454508.738           1.336300
160             83.450              2457917.475           0.909000
161             87.045              2458425.789           0.026759

     Transit Duration [hours]  Radial Velocity Amplitude [m/s]
0                    1.543900                            6.020
1                    3.240000                            3.800
2                    4.290000                            4.696
3                    2.387000                          590.000
4                    1.693000                            3.220
..                        ...                              ...
157                  2.426400                           62.000
158                  0.900451                            3.540
159                  2.683900                           90.170
160                  2.772000                           80.500
161                  2.952024                            1.185

[162 rows x 18 columns]
```

In [26]:

```python
#check for missing values or Na in each column,
#True for missing, false for NOT missing

missing_values = new_planet_data.isna()

#adds the number of rows in each column with missing values
row_total=missing_values.sum()
print(row_total)
print('________________________Exoplanets________________________')

missing_values = stars_data.isna()

#adds the number of rows in each column with missing values
row_total=missing_values.sum()
print(row_total)
print('________________________Stars_____________________________')


missing_values = system_data.isna()
```

```
row_total=missing_values.sum()
print(row_total)
print('______________________Solar System________________________')


missing_values = discoveries_data.isna()

row_total=missing_values.sum()
print(row_total)
print('______________________Discoveries_________________________')
```

```
Host Name                                   0
Planet Name                                 0
Planet Letter                               0
Number of Moons                             0
Orbital Period [days]                       0
Orbit Semi-Major Axis [au]                  0
Angular Separation [mas]                    0
Planet Radius [Earth Radius]                0
Planet Mass or Mass*sin(i) [Earth Mass]     0
Planet Density [g/cm**3]                    0
Eccentricity                                0
Insolation Flux [Earth Flux]                0
Equilibrium Temperature [K]                 0
Inclination [deg]                           0
Transit Midpoint [days]                     0
Transit Depth [%]                           0
Transit Duration [hours]                    0
Radial Velocity Amplitude [m/s]             0
dtype: int64
________________________Exoplanets________________________
Host Name                                     0
System ID                                     0
Spectral Type                                 0
Stellar Radius [Solar Radius]                 0
Stellar Mass [Solar mass]                     0
Stellar Metallicity [dex]                     0
Stellar Luminosity [log(Solar)]               0
Stellar Surface Gravity [log10(cm/s**2)]      0
Stellar Age [Gyr]                             0
Stellar Density [g/cm**3]                     0
Stellar Rotational Velocity [km/s]            0
Stellar Rotational Period [days]              0
Systemic Radial Velocity [km/s]               0
dtype: int64
________________________Stars_____________________________
Host Name              0
Number of Stars        0
Number of Planets      0
RA [deg]               0
Dec [deg]              0
Distance [pc]          0
dtype: int64
______________________Solar System________________________
Host Name               0
Discovery Method        0
Discovery Year          0
Discovery Facility      0
dtype: int64
______________________Discoveries_________________________
```

In [27]:
```
#count of entries before drop_duplicates
star_entries_before = len(stars_data)
print( 'stars before: ', star_entries_before)

stars_data.drop_duplicates(subset=['Spectral Type', 'Stellar Radius [Solar Radius]'
'Stellar Mass [Solar mass]', 'Stellar Metallicity [dex]',
```

```
'Stellar Luminosity [log(Solar)]', 'Stellar Surface Gravity [log10(cm/s**2)]',
'Stellar Age [Gyr]', 'Stellar Density [g/cm**3]', 'Stellar Rotational Velocity [km/
'Stellar Rotational Period [days]', 'Systemic Radial Velocity [km/s]'], inplace=Tru

#after drops
star_entries_after = len(stars_data)
print( 'stars after:', star_entries_after)

#difference
star_entries_dropped = star_entries_before - star_entries_after
print("Duplicates Removed:", star_entries_dropped)
```

```
stars before:  162
stars after: 131
Duplicates Removed: 31
```

In [28]:

```
#set variable number of rows in dataset
number_rows = len(stars_data)

## add new columns using data.insert(location, column name, rows) at the end of the
stars_data.insert(0, 'Star ID', range(1,  number_rows + 1))

print(stars_data)
```

```
     Star ID Host Name  System ID Spectral Type  \
0          1    55 Cnc          1          G8 V   
1          2    AU Mic          2          M1 V   
3          3  CoRoT-18          4          G9 V   
4          4    G 9-40          5        M2.5 V   
5          5   GJ 1132          6        M4.5 V   
..       ...       ...        ...           ...   
157      127   WASP-96        158            G8   
158      128  Wolf 327        159        M2.5 V   
159      129    XO-2 N        160          K0 V   
160      130      XO-7        161          G0 V   
161      131  HD 39091        162          G0 V   

     Stellar Radius [Solar Radius]  Stellar Mass [Solar mass]  \
0                           0.9430                     0.9050   
1                           0.8620                     0.6350   
3                           1.0000                     0.9500   
4                           0.3026                     0.2952   
5                           0.2211                     0.1945   
..                             ...                        ...   
157                         1.0500                     1.0600   
158                         0.4060                     0.4050   
159                         0.9900                     0.9710   
160                         1.4800                     1.4050   
161                         1.1700                     1.0700   

     Stellar Metallicity [dex]  Stellar Luminosity [log(Solar)]  \
0                        0.350                         -0.19695   
1                        0.010                         -0.97757   
3                       -0.100                         -0.17000   
4                       -0.070                         -1.96019   
5                       -0.170                         -2.32148   
..                         ...                              ...   
157                      0.140                          0.03226   
158                     -0.170                         -1.62986   
159                      0.430                         -0.08823   
160                      0.432                          0.43132   
161                      0.090                          0.15836   

     Stellar Surface Gravity [log10(cm/s**2)]  Stellar Age [Gyr]  \
0                                       4.430              5.500   
1                                       4.370              0.019   
3                                       4.400              0.600   
4                                       4.840              9.900   
5                                       5.037              5.000   
..                                        ...                ...   
157                                     4.420              8.000   
158                                     4.870              4.100   
159                                     4.433              7.800   
160                                     4.246              1.180   
161                                     4.430              2.980   

     Stellar Density [g/cm**3]  Stellar Rotational Velocity [km/s]  \
0                     1.521203                                1.23   
1                     1.750000                                9.20   
3                     1.350000                                8.00   
```

```
4                    1.620000                               2.00
5                   25.300000                               2.00
..                        ...                                ...
157                  1.299000                               1.50
158                  8.540000                               2.00
159                  1.410000                               1.07
160                  0.451084                               6.00
161                  0.944584                               3.16

     Stellar Rotational Period [days]  Systemic Radial Velocity [km/s]
0                               38.80                         27.40949
1                                4.86                         -6.31031
3                                5.40                         29.55000
4                               30.00                         14.65000
5                              122.30                         35.07880
..                                ...                              ...
157                             20.16                         -1.23300
158                             44.40                          9.39000
159                             28.60                         46.93290
160                              3.62                        -12.98300
161                             18.30                         10.70916

[131 rows x 14 columns]
```

In [29]:

```
#count of entries before drop_duplicates
system_entries_before = len(system_data)
print('system before: ', system_entries_before)

system_data.drop_duplicates(subset=['Host Name'], inplace=True)

#after
system_entries_after = len(system_data)
print( 'system after:', system_entries_after)

#difference
system_entries_dropped = system_entries_before - system_entries_after

print("Duplicates Removed:", system_entries_dropped)
```

```
system before:  162
system after: 130
Duplicates Removed: 32
```

In [30]:

```
#set variable number of rows in dataset
number_rows = len(system_data)

## add new columns using data.insert(location, column name, rows) at the end of the
system_data.insert(0, 'System ID', range(1,  number_rows + 1))

print(system_data)
```

```
     System ID Host Name  Number of Stars  Number of Planets    RA [deg]  \
0            1    55 Cnc                2                  7  133.146837
1            2    AU Mic                1                  4  311.291137
3            3  CoRoT-18                1                  1   98.172417
4            4    G 9-40                1                  1  134.718851
5            5   GJ 1132                1                  2  153.709070
..         ...       ...              ...                ...         ...
157        126   WASP-96                1                  1    1.046570
158        127  Wolf 327                1                  1  148.378320
159        128    XO-2 N                2                  3  117.026769
160        129      XO-7                1                  1  277.478075
161        130  HD 39091                1                  3   84.299280

     Dec [deg]  Distance [pc]
0    28.329815        12.5855
1   -31.342450         9.7221
3    -0.031605       764.8890
4    21.074796        27.9276
5   -47.154936        12.6130
..         ...            ...
157 -47.360635       352.4640
158  35.570548        28.5321
159  50.225147       154.2730
160  85.233321       234.1490
161 -80.464604        18.2702

[130 rows x 7 columns]
```

In [31]:

```python
#count of entries before drop_duplicates
discovery_entries_before = len(discoveries_data)
print('discoveries before: ', discovery_entries_before)

discoveries_data.drop_duplicates(subset=['Discovery Method', 'Discovery Year', 'Dis

#after
discovery_entries_after = len(discoveries_data)
print( 'discoveries after:', discovery_entries_after)

#difference
discovery_entries_dropped = discovery_entries_before - discovery_entries_after

print("Duplicates Removed:", discovery_entries_dropped)
```

```
discoveries before:  162
discoveries after: 56
Duplicates Removed: 106
```

In [32]:

```python
#set variable number of rows in dataset
number_rows = len(discoveries_data)

## add new columns using data.insert(location, column name, rows) at the end of the
discoveries_data.insert(0, 'Discovery ID', range(1,  number_rows + 1))

print(discoveries_data)
```

```
     Discovery ID   Host Name Discovery Method  Discovery Year  \
0               1      55 Cnc  Radial Velocity          2004.0   
1               2      AU Mic          Transit          2020.0   
2               3      AU Mic          Transit          2021.0   
3               4    CoRoT-18          Transit          2011.0   
4               5      G 9-40          Transit          2019.0   
5               6     GJ 1132          Transit          2015.0   
6               7     GJ 1214          Transit          2009.0   
7               8     GJ 3929          Transit          2022.0   
8               9      GJ 436  Radial Velocity          2004.0   
9              10      GJ 486  Radial Velocity          2021.0   
10             11     GJ 9827          Transit          2017.0   
13             12   Gliese 12          Transit          2024.0   
14             13    HAT-P-11          Transit          2008.0   
15             14    HAT-P-19          Transit          2010.0   
19             15     HAT-P-3          Transit          2007.0   
20             16     HATS-16          Transit          2016.0   
21             17     HATS-18          Transit          2016.0   
24             18   HD 118203  Radial Velocity          2005.0   
25             19   HD 136352  Radial Velocity          2019.0   
28             20    HD 15337          Transit          2019.0   
30             21    HD 17156  Radial Velocity          2007.0   
34             22   HD 209458  Radial Velocity          1999.0   
38             23     HD 3167          Transit          2016.0   
42             24    HD 73344          Transit          2024.0   
56             25      K2-141          Transit          2018.0   
58             26       K2-19          Transit          2015.0   
66             27     KELT-18          Transit          2017.0   
67             28  Kepler-103          Transit          2014.0   
69             29   Kepler-17          Transit          2011.0   
70             30   Kepler-21          Transit          2012.0   
74             31  Kepler-539          Transit          2016.0   
75             32   Kepler-63          Transit          2013.0   
85             33     Qatar-3          Transit          2017.0   
89             34     Qatar-7          Transit          2019.0   
101            35    TOI-1416          Transit          2023.0   
105            36    TOI-2093          Transit          2025.0   
127            37     WASP-10          Transit          2008.0   
128            38    WASP-107          Transit          2017.0   
129            39    WASP-113          Transit          2016.0   
130            40    WASP-118          Transit          2016.0   
132            41    WASP-124          Transit          2016.0   
135            42    WASP-148          Transit          2020.0   
136            43    WASP-166          Transit          2019.0   
137            44    WASP-172          Transit          2018.0   
139            45  WASP-180 A          Transit          2019.0   
141            46    WASP-186          Transit          2020.0   
142            47     WASP-19          Transit          2009.0   
143            48     WASP-22          Transit          2010.0   
144            49      WASP-4          Transit          2007.0   
145            50     WASP-43          Transit          2011.0   
147            51     WASP-47          Transit          2012.0   
151            52     WASP-69          Transit          2014.0   
154            53     WASP-89          Transit          2015.0   
159            54     XO-2 N          Transit          2007.0   
160            55       XO-7          Transit          2019.0   
```

161	56	HD 39091	Transit	2018.0

	Discovery Facility
0	McDonald Observatory
1	Transiting Exoplanet Survey Satellite (TESS)
2	Transiting Exoplanet Survey Satellite (TESS)
3	CoRoT
4	K2
5	Cerro Tololo Inter-American Observatory
6	MEarth Project
7	Transiting Exoplanet Survey Satellite (TESS)
8	W. M. Keck Observatory
9	Calar Alto Observatory
10	K2
13	Transiting Exoplanet Survey Satellite (TESS)
14	HATNet
15	HATNet
19	HATNet
20	HATSouth
21	Multiple Observatories
24	Haute-Provence Observatory
25	La Silla Observatory
28	Transiting Exoplanet Survey Satellite (TESS)
30	Subaru Telescope
34	W. M. Keck Observatory
38	K2
42	K2
56	K2
58	K2
66	KELT-North
67	Kepler
69	Kepler
70	Kepler
74	Kepler
75	Kepler
85	Qatar
89	Qatar
101	Transiting Exoplanet Survey Satellite (TESS)
105	Transiting Exoplanet Survey Satellite (TESS)
127	SuperWASP
128	SuperWASP-South
129	SuperWASP-North
130	SuperWASP
132	SuperWASP-South
135	SuperWASP-North
136	WASP-South
137	SuperWASP-South
139	SuperWASP
141	SuperWASP
142	SuperWASP
143	SuperWASP
144	SuperWASP
145	SuperWASP
147	SuperWASP
151	SuperWASP
154	SuperWASP

```
159                                              XO
160                                              XO
161  Transiting Exoplanet Survey Satellite (TESS)
```

In [33]:
```
#count of entries before drop_duplicates
planet_entries_before = len(new_planet_data)
print('planets before: ', planet_entries_before)

new_planet_data.drop_duplicates(subset=['Planet Name'], inplace=True)

#after
planet_entries_after = len(new_planet_data)
print( 'planet after:', planet_entries_after)

#difference
planet_entries_dropped = planet_entries_before - planet_entries_after

print("Duplicates Removed:", planet_entries_dropped)
```

```
planets before:  162
planet after: 162
Duplicates Removed: 0
```

In [34]:
```
#set variable number of rows in dataset
number_rows = len(new_planet_data)

new_planet_data.insert(0,'Planet ID', range(1, number_rows+1))

print(new_planet_data)
```

```
     Planet ID Host Name Planet Name Planet Letter  Number of Moons  \
0            1    55 Cnc    55 Cnc e             e                0
1            2    AU Mic    AU Mic b             b                0
2            3    AU Mic    AU Mic c             c                0
3            4  CoRoT-18  CoRoT-18 b             b                0
4            5    G 9-40    G 9-40 b             b                0
..         ...       ...         ...           ...              ...
157        158   WASP-96   WASP-96 b             b                0
158        159  Wolf 327  Wolf 327 b             b                0
159        160    XO-2 N    XO-2 N b             b                0
160        161      XO-7      XO-7 b             b                0
161        162  HD 39091    pi Men c             c                0

     Orbital Period [days]  Orbit Semi-Major Axis [au]  \
0                 0.736547                     0.01544
1                 8.463446                     0.07000
2                18.859023                     0.11900
3                 1.900069                     0.02950
4                 5.745998                     0.04180
..                     ...                         ...
157               3.425260                     0.04530
158               0.573474                     0.01000
159               2.615862                     0.03680
160               2.864142                     0.04421
161               6.267840                     0.06839

     Angular Separation [mas]  Planet Radius [Earth Radius]  \
0                      1.2300                      1.875000
1                      7.2000                      4.790000
2                     12.2000                      2.790000
3                      0.0386                     14.680000
4                      1.5000                      1.900000
..                        ...                           ...
157                    0.1290                     13.450000
158                    0.3500                      1.240000
159                    0.2390                     11.130537
160                    0.1890                     15.389957
161                    3.7400                      2.018900

     Planet Mass or Mass*sin(i) [Earth Mass]  Planet Density [g/cm**3]  \
0                                    7.99000                     6.660
1                                    8.99000                     0.490
2                                   14.46000                     3.900
3                                 1102.82000                     2.200
4                                    4.00000                     3.200
..                                       ...                       ...
157                                152.55000                     0.370
158                                  2.53000                     7.240
159                                179.89178                     0.715
160                                225.34147                     0.340
161                                  3.63000                     2.800

     Eccentricity  Insolation Flux [Earth Flux]  Equilibrium Temperature [K]  \
0           0.050                     2657.8322                       1958.0
1           0.070                       21.2000                        554.8
2           0.180                        7.3000                        424.7
```

```
3          0.080                    823.7009                        1550.0
4          0.000                      6.2700                         440.6
..           ...                         ...                           ...
157        0.000                    517.0000                        1285.0
158        0.000                    233.9000                         996.0
159        0.028                    428.0360                        1328.0
160        0.038                   1540.0000                        1743.0
161        0.000                    309.0000                        1169.8

     Inclination [deg]  Transit Midpoint [days]  Transit Depth [%]  \
0               83.590              2457063.210           0.038467   
1               88.390              2458330.351           0.249000   
2               89.460              2458342.223           0.085000   
3               86.500              2455321.724           2.571701   
4               89.030              2459503.327           0.365000   
..                 ...                      ...                ...   
157             85.600              2456258.062           1.380000   
158             84.890              2459252.981           0.081839   
159             88.010              2454508.738           1.336300   
160             83.450              2457917.475           0.909000   
161             87.045              2458425.789           0.026759   

     Transit Duration [hours]  Radial Velocity Amplitude [m/s]  
0                    1.543900                            6.020  
1                    3.240000                            3.800  
2                    4.290000                            4.696  
3                    2.387000                          590.000  
4                    1.693000                            3.220  
..                        ...                              ...  
157                  2.426400                           62.000  
158                  0.900451                            3.540  
159                  2.683900                           90.170  
160                  2.772000                           80.500  
161                  2.952024                            1.185  

[162 rows x 19 columns]
```

In [35]:

```python
#check for missing values or Na in each column,
#True for missing, false for NOT missing

missing_values = new_planet_data.isna()

#adds the number of rows in each column with missing values
row_total=missing_values.sum()
print(row_total)
print('________________________Exoplanets________________________')

missing_values = stars_data.isna()

#adds the number of rows in each column with missing values
row_total=missing_values.sum()
print(row_total)
print('________________________Stars_____________________________')


missing_values = system_data.isna()
```

```
row_total=missing_values.sum()
print(row_total)
print('______________________Solar System________________________')


missing_values = discoveries_data.isna()

row_total=missing_values.sum()
print(row_total)
print('______________________Discoveries_________________________')
```

```
Planet ID                                   0
Host Name                                   0
Planet Name                                 0
Planet Letter                               0
Number of Moons                             0
Orbital Period [days]                       0
Orbit Semi-Major Axis [au]                  0
Angular Separation [mas]                    0
Planet Radius [Earth Radius]                0
Planet Mass or Mass*sin(i) [Earth Mass]     0
Planet Density [g/cm**3]                    0
Eccentricity                                0
Insolation Flux [Earth Flux]                0
Equilibrium Temperature [K]                 0
Inclination [deg]                           0
Transit Midpoint [days]                     0
Transit Depth [%]                           0
Transit Duration [hours]                    0
Radial Velocity Amplitude [m/s]             0
dtype: int64
________________________Exoplanets________________________
Star ID                                      0
Host Name                                    0
System ID                                    0
Spectral Type                                0
Stellar Radius [Solar Radius]                0
Stellar Mass [Solar mass]                    0
Stellar Metallicity [dex]                    0
Stellar Luminosity [log(Solar)]              0
Stellar Surface Gravity [log10(cm/s**2)]     0
Stellar Age [Gyr]                            0
Stellar Density [g/cm**3]                    0
Stellar Rotational Velocity [km/s]           0
Stellar Rotational Period [days]             0
Systemic Radial Velocity [km/s]              0
dtype: int64
________________________Stars_____________________________
System ID             0
Host Name             0
Number of Stars       0
Number of Planets     0
RA [deg]              0
Dec [deg]             0
Distance [pc]         0
dtype: int64
______________________Solar System________________________
Discovery ID           0
Host Name              0
Discovery Method       0
Discovery Year         0
Discovery Facility     0
dtype: int64
______________________Discoveries_________________________
```

In [36]:

```
print(new_planet_data)
```

```
     Planet ID Host Name Planet Name Planet Letter  Number of Moons  \
0            1    55 Cnc    55 Cnc e             e                0   
1            2    AU Mic    AU Mic b             b                0   
2            3    AU Mic    AU Mic c             c                0   
3            4  CoRoT-18  CoRoT-18 b             b                0   
4            5    G 9-40    G 9-40 b             b                0   
..         ...       ...         ...           ...              ...   
157        158   WASP-96   WASP-96 b             b                0   
158        159  Wolf 327  Wolf 327 b             b                0   
159        160    XO-2 N    XO-2 N b             b                0   
160        161      XO-7      XO-7 b             b                0   
161        162  HD 39091    pi Men c             c                0   

     Orbital Period [days]  Orbit Semi-Major Axis [au]  \
0                 0.736547                     0.01544   
1                 8.463446                     0.07000   
2                18.859023                     0.11900   
3                 1.900069                     0.02950   
4                 5.745998                     0.04180   
..                     ...                         ...   
157               3.425260                     0.04530   
158               0.573474                     0.01000   
159               2.615862                     0.03680   
160               2.864142                     0.04421   
161               6.267840                     0.06839   

     Angular Separation [mas]  Planet Radius [Earth Radius]  \
0                      1.2300                      1.875000   
1                      7.2000                      4.790000   
2                     12.2000                      2.790000   
3                      0.0386                     14.680000   
4                      1.5000                      1.900000   
..                        ...                           ...   
157                    0.1290                     13.450000   
158                    0.3500                      1.240000   
159                    0.2390                     11.130537   
160                    0.1890                     15.389957   
161                    3.7400                      2.018900   

     Planet Mass or Mass*sin(i) [Earth Mass]  Planet Density [g/cm**3]  \
0                                    7.99000                     6.660   
1                                    8.99000                     0.490   
2                                   14.46000                     3.900   
3                                 1102.82000                     2.200   
4                                    4.00000                     3.200   
..                                       ...                       ...   
157                                152.55000                     0.370   
158                                  2.53000                     7.240   
159                                179.89178                     0.715   
160                                225.34147                     0.340   
161                                  3.63000                     2.800   

     Eccentricity  Insolation Flux [Earth Flux]  Equilibrium Temperature [K]  \
0           0.050                     2657.8322                       1958.0   
1           0.070                       21.2000                        554.8   
2           0.180                        7.3000                        424.7   
```

```
3          0.080                     823.7009                        1550.0
4          0.000                       6.2700                         440.6
..           ...                          ...                           ...
157        0.000                     517.0000                        1285.0
158        0.000                     233.9000                         996.0
159        0.028                     428.0360                        1328.0
160        0.038                    1540.0000                        1743.0
161        0.000                     309.0000                        1169.8

     Inclination [deg]  Transit Midpoint [days]  Transit Depth [%]  \
0               83.590              2457063.210           0.038467
1               88.390              2458330.351           0.249000
2               89.460              2458342.223           0.085000
3               86.500              2455321.724           2.571701
4               89.030              2459503.327           0.365000
..                 ...                      ...                ...
157             85.600              2456258.062           1.380000
158             84.890              2459252.981           0.081839
159             88.010              2454508.738           1.336300
160             83.450              2457917.475           0.909000
161             87.045              2458425.789           0.026759

     Transit Duration [hours]  Radial Velocity Amplitude [m/s]
0                    1.543900                            6.020
1                    3.240000                            3.800
2                    4.290000                            4.696
3                    2.387000                          590.000
4                    1.693000                            3.220
..                        ...                              ...
157                  2.426400                           62.000
158                  0.900451                            3.540
159                  2.683900                           90.170
160                  2.772000                           80.500
161                  2.952024                            1.185

[162 rows x 19 columns]
```

In [37]:

```
#merging or linking tables Stars with Systems before MySQL Lab, one to Many
# we can use map() and dict() because system_data is already normalized
#and each row has a unique system or host name

#AUTOMATICALLY build the mapping dictionary from your stars DataFrame

#This pairs every star name with its MySQL-ready ID, no matter how many hundreds th

system_id_lookup = dict(zip(system_data['Host Name'], system_data['System ID']))

#Use .map() to fill the entire 'star_id' column at once

stars_data['System ID'] = stars_data['Host Name'].map(system_id_lookup)

#Clean up: Drop the text star_name column so MySQL stays perfectly relational

print(stars_data)
```

	Star ID	Host Name	System ID	Spectral Type	\
0	1	55 Cnc	1	G8 V	
1	2	AU Mic	2	M1 V	
3	3	CoRoT-18	3	G9 V	
4	4	G 9-40	4	M2.5 V	
5	5	GJ 1132	5	M4.5 V	
..	...	...	...	...	
157	127	WASP-96	126	G8	
158	128	Wolf 327	127	M2.5 V	
159	129	XO-2 N	128	K0 V	
160	130	XO-7	129	G0 V	
161	131	HD 39091	130	G0 V	

	Stellar Radius [Solar Radius]	Stellar Mass [Solar mass]	\
0	0.9430	0.9050	
1	0.8620	0.6350	
3	1.0000	0.9500	
4	0.3026	0.2952	
5	0.2211	0.1945	
..	...	...	
157	1.0500	1.0600	
158	0.4060	0.4050	
159	0.9900	0.9710	
160	1.4800	1.4050	
161	1.1700	1.0700	

	Stellar Metallicity [dex]	Stellar Luminosity [log(Solar)]	\
0	0.350	-0.19695	
1	0.010	-0.97757	
3	-0.100	-0.17000	
4	-0.070	-1.96019	
5	-0.170	-2.32148	
..	...	...	
157	0.140	0.03226	
158	-0.170	-1.62986	
159	0.430	-0.08823	
160	0.432	0.43132	
161	0.090	0.15836	

	Stellar Surface Gravity [log10(cm/s**2)]	Stellar Age [Gyr]	\
0	4.430	5.500	
1	4.370	0.019	
3	4.400	0.600	
4	4.840	9.900	
5	5.037	5.000	
..	...	...	
157	4.420	8.000	
158	4.870	4.100	
159	4.433	7.800	
160	4.246	1.180	
161	4.430	2.980	

	Stellar Density [g/cm**3]	Stellar Rotational Velocity [km/s]	\
0	1.521203	1.23	
1	1.750000	9.20	
3	1.350000	8.00	

```
4                   1.620000                         2.00
5                  25.300000                         2.00
..                       ...                          ...
157                 1.299000                         1.50
158                 8.540000                         2.00
159                 1.410000                         1.07
160                 0.451084                         6.00
161                 0.944584                         3.16

     Stellar Rotational Period [days]  Systemic Radial Velocity [km/s]
0                               38.80                         27.40949
1                                4.86                         -6.31031
3                                5.40                         29.55000
4                               30.00                         14.65000
5                              122.30                         35.07880
..                                ...                              ...
157                             20.16                         -1.23300
158                             44.40                          9.39000
159                             28.60                         46.93290
160                              3.62                        -12.98300
161                             18.30                         10.70916

[131 rows x 14 columns]
```

In [38]:

```
#merging or linking tables Stars and Exoplanets before MySQL Lab
# creating junction data sets for junction data tables, Many to Many

junction = pd.merge(
    new_planet_data,
    stars_data,
    on = 'Host Name',
    how= 'inner'
)

Star_Exoplanet_Orbits = junction[['Star ID','Planet ID']]

print(Star_Exoplanet_Orbits)
```

```
     Star ID  Planet ID
0          1          1
1          2          2
2          2          3
3          3          4
4          4          5
..       ...        ...
160      127        158
161      128        159
162      129        160
163      130        161
164      131        162

[165 rows x 2 columns]
```

In [39]:

```
#merging or linking tables Exoplanet and Discoveries before MySQL Lab
# creating junction data sets for junction data tables, Many to Many
```

```
junction = pd.merge(
    new_planet_data,
    discoveries_data,
    on = 'Host Name',
    how= 'inner'
)

Exoplanet_Discoveries = junction[['Discovery ID','Planet ID']]


print(Exoplanet_Discoveries)
```

```
    Discovery ID  Planet ID
0              1          1
1              2          2
2              3          2
3              2          3
4              3          3
..           ...        ...
64            52        152
65            53        155
66            54        160
67            55        161
68            56        162

[69 rows x 2 columns]
```

In [40]:
```
#removing Host Name from all datasets where it's not needed for normalization

stars_data = stars_data.drop(columns=['Host Name'])
new_planet_data = new_planet_data.drop(columns=['Host Name'])
discoveries_data = discoveries_data.drop(columns=['Host Name'])
```

In [41]:
```
#store cleaned datasets in new CSV files

#save stars data
cleaned_stars_data = stars_data.copy()
cleaned_stars_data.to_csv('cleaned_stars_data.csv', index=False)
print(cleaned_stars_data)

#save solar system data
cleaned_system_data = system_data.copy()
cleaned_system_data.to_csv('cleaned_system_data.csv', index=False)
print(cleaned_system_data)

#save discovery data
cleaned_discoveries_data = discoveries_data.copy()
cleaned_discoveries_data.to_csv('cleaned_discoveries_data.csv', index=False)
print(cleaned_discoveries_data)

#store cleaned exoplanet data
cleaned_planet_data = new_planet_data.copy()
cleaned_planet_data.to_csv('cleaned_planet_data.csv', index=False)
print(cleaned_planet_data)

#junction table Star_Exoplanet_Orbits
```

```
Star_Exoplanet_Orbits_cleaned = Star_Exoplanet_Orbits.copy()
Star_Exoplanet_Orbits_cleaned.to_csv('Star_Exoplanet_Orbits_cleaned.csv', index=Fal
print(Star_Exoplanet_Orbits_cleaned)

#junction table Exoplanet Discoveries
Exoplanet_Discoveries_cleaned = Exoplanet_Discoveries.copy()
Exoplanet_Discoveries_cleaned.to_csv('Exoplanet_Discoveries_cleaned.csv', index=Fal
print(Exoplanet_Discoveries_cleaned)
```

	Star ID	System ID	Spectral Type	Stellar Radius [Solar Radius] \
0	1	1	G8 V	0.9430
1	2	2	M1 V	0.8620
3	3	3	G9 V	1.0000
4	4	4	M2.5 V	0.3026
5	5	5	M4.5 V	0.2211
..	...	...	...	...
157	127	126	G8	1.0500
158	128	127	M2.5 V	0.4060
159	129	128	K0 V	0.9900
160	130	129	G0 V	1.4800
161	131	130	G0 V	1.1700

	Stellar Mass [Solar mass]	Stellar Metallicity [dex] \
0	0.9050	0.350
1	0.6350	0.010
3	0.9500	-0.100
4	0.2952	-0.070
5	0.1945	-0.170
..	...	...
157	1.0600	0.140
158	0.4050	-0.170
159	0.9710	0.430
160	1.4050	0.432
161	1.0700	0.090

	Stellar Luminosity [log(Solar)] \
0	-0.19695
1	-0.97757
3	-0.17000
4	-1.96019
5	-2.32148
..	...
157	0.03226
158	-1.62986
159	-0.08823
160	0.43132
161	0.15836

	Stellar Surface Gravity [log10(cm/s**2)]	Stellar Age [Gyr] \
0	4.430	5.500
1	4.370	0.019
3	4.400	0.600
4	4.840	9.900
5	5.037	5.000
..	...	...
157	4.420	8.000
158	4.870	4.100
159	4.433	7.800
160	4.246	1.180
161	4.430	2.980

	Stellar Density [g/cm**3]	Stellar Rotational Velocity [km/s] \
0	1.521203	1.23
1	1.750000	9.20
3	1.350000	8.00

```
4                   1.620000                                2.00  
5                  25.300000                                2.00  
..                       ...                                 ...  
157                 1.299000                                1.50  
158                 8.540000                                2.00  
159                 1.410000                                1.07  
160                 0.451084                                6.00  
161                 0.944584                                3.16  

     Stellar Rotational Period [days]  Systemic Radial Velocity [km/s]  
0                               38.80                         27.40949  
1                                4.86                         -6.31031  
3                                5.40                         29.55000  
4                               30.00                         14.65000  
5                              122.30                         35.07880  
..                                ...                              ...  
157                             20.16                         -1.23300  
158                             44.40                          9.39000  
159                             28.60                         46.93290  
160                              3.62                        -12.98300  
161                             18.30                         10.70916  

[131 rows x 13 columns]
     System ID Host Name  Number of Stars  Number of Planets    RA [deg]  \
0            1    55 Cnc                2                  7  133.146837   
1            2    AU Mic                1                  4  311.291137   
3            3  CoRoT-18                1                  1   98.172417   
4            4    G 9-40                1                  1  134.718851   
5            5   GJ 1132                1                  2  153.709070   
..         ...       ...              ...                ...         ...   
157        126   WASP-96                1                  1    1.046570   
158        127  Wolf 327                1                  1  148.378320   
159        128    XO-2 N                2                  3  117.026769   
160        129      XO-7                1                  1  277.478075   
161        130  HD 39091                1                  3   84.299280   

     Dec [deg]  Distance [pc]  
0    28.329815        12.5855  
1   -31.342450         9.7221  
3    -0.031605       764.8890  
4    21.074796        27.9276  
5   -47.154936        12.6130  
..         ...            ...  
157 -47.360635       352.4640  
158  35.570548        28.5321  
159  50.225147       154.2730  
160  85.233321       234.1490  
161 -80.464604        18.2702  

[130 rows x 7 columns]
     Discovery ID Discovery Method  Discovery Year  \
0               1  Radial Velocity          2004.0   
1               2          Transit          2020.0   
2               3          Transit          2021.0   
3               4          Transit          2011.0   
4               5          Transit          2019.0   
```

```
5                6          Transit          2015.0
6                7          Transit          2009.0
7                8          Transit          2022.0
8                9  Radial Velocity          2004.0
9               10  Radial Velocity          2021.0
10              11          Transit          2017.0
13              12          Transit          2024.0
14              13          Transit          2008.0
15              14          Transit          2010.0
19              15          Transit          2007.0
20              16          Transit          2016.0
21              17          Transit          2016.0
24              18  Radial Velocity          2005.0
25              19  Radial Velocity          2019.0
28              20          Transit          2019.0
30              21  Radial Velocity          2007.0
34              22  Radial Velocity          1999.0
38              23          Transit          2016.0
42              24          Transit          2024.0
56              25          Transit          2018.0
58              26          Transit          2015.0
66              27          Transit          2017.0
67              28          Transit          2014.0
69              29          Transit          2011.0
70              30          Transit          2012.0
74              31          Transit          2016.0
75              32          Transit          2013.0
85              33          Transit          2017.0
89              34          Transit          2019.0
101             35          Transit          2023.0
105             36          Transit          2025.0
127             37          Transit          2008.0
128             38          Transit          2017.0
129             39          Transit          2016.0
130             40          Transit          2016.0
132             41          Transit          2016.0
135             42          Transit          2020.0
136             43          Transit          2019.0
137             44          Transit          2018.0
139             45          Transit          2019.0
141             46          Transit          2020.0
142             47          Transit          2009.0
143             48          Transit          2010.0
144             49          Transit          2007.0
145             50          Transit          2011.0
147             51          Transit          2012.0
151             52          Transit          2014.0
154             53          Transit          2015.0
159             54          Transit          2007.0
160             55          Transit          2019.0
161             56          Transit          2018.0

                              Discovery Facility
0                           McDonald Observatory
1    Transiting Exoplanet Survey Satellite (TESS)
2    Transiting Exoplanet Survey Satellite (TESS)
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3                                               CoRoT
4                                                  K2
5         Cerro Tololo Inter-American Observatory
6                                     MEarth Project
7    Transiting Exoplanet Survey Satellite (TESS)
8                            W. M. Keck Observatory
9                            Calar Alto Observatory
10                                                 K2
13   Transiting Exoplanet Survey Satellite (TESS)
14                                             HATNet
15                                             HATNet
19                                             HATNet
20                                           HATSouth
21                           Multiple Observatories
24                       Haute-Provence Observatory
25                             La Silla Observatory
28   Transiting Exoplanet Survey Satellite (TESS)
30                                  Subaru Telescope
34                           W. M. Keck Observatory
38                                                 K2
42                                                 K2
56                                                 K2
58                                                 K2
66                                         KELT-North
67                                             Kepler
69                                             Kepler
70                                             Kepler
74                                             Kepler
75                                             Kepler
85                                              Qatar
89                                              Qatar
101  Transiting Exoplanet Survey Satellite (TESS)
105  Transiting Exoplanet Survey Satellite (TESS)
127                                         SuperWASP
128                                   SuperWASP-South
129                                   SuperWASP-North
130                                         SuperWASP
132                                   SuperWASP-South
135                                   SuperWASP-North
136                                        WASP-South
137                                   SuperWASP-South
139                                         SuperWASP
141                                         SuperWASP
142                                         SuperWASP
143                                         SuperWASP
144                                         SuperWASP
145                                         SuperWASP
147                                         SuperWASP
151                                         SuperWASP
154                                         SuperWASP
159                                                XO
160                                                XO
161  Transiting Exoplanet Survey Satellite (TESS)
     Planet ID Planet Name Planet Letter  Number of Moons  \
0            1    55 Cnc e             e                0
1            2    AU Mic b             b                0
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2            3    AU Mic c            c              0
3            4  CoRoT-18 b            b              0
4            5    G 9-40 b            b              0
..         ...         ...          ...            ...
157        158   WASP-96 b            b              0
158        159  Wolf 327 b            b              0
159        160    XO-2 N b            b              0
160        161      XO-7 b            b              0
161        162    pi Men c            c              0

     Orbital Period [days]  Orbit Semi-Major Axis [au]  \
0                 0.736547                     0.01544   
1                 8.463446                     0.07000   
2                18.859023                     0.11900   
3                 1.900069                     0.02950   
4                 5.745998                     0.04180   
..                     ...                         ...   
157               3.425260                     0.04530   
158               0.573474                     0.01000   
159               2.615862                     0.03680   
160               2.864142                     0.04421   
161               6.267840                     0.06839   

     Angular Separation [mas]  Planet Radius [Earth Radius]  \
0                      1.2300                      1.875000   
1                      7.2000                      4.790000   
2                     12.2000                      2.790000   
3                      0.0386                     14.680000   
4                      1.5000                      1.900000   
..                        ...                           ...   
157                    0.1290                     13.450000   
158                    0.3500                      1.240000   
159                    0.2390                     11.130537   
160                    0.1890                     15.389957   
161                    3.7400                      2.018900   

     Planet Mass or Mass*sin(i) [Earth Mass]  Planet Density [g/cm**3]  \
0                                    7.99000                     6.660   
1                                    8.99000                     0.490   
2                                   14.46000                     3.900   
3                                 1102.82000                     2.200   
4                                    4.00000                     3.200   
..                                       ...                       ...   
157                                152.55000                     0.370   
158                                  2.53000                     7.240   
159                                179.89178                     0.715   
160                                225.34147                     0.340   
161                                  3.63000                     2.800   

     Eccentricity  Insolation Flux [Earth Flux]  Equilibrium Temperature [K]  \
0           0.050                     2657.8322                       1958.0   
1           0.070                       21.2000                        554.8   
2           0.180                        7.3000                        424.7   
3           0.080                      823.7009                       1550.0   
4           0.000                        6.2700                        440.6   
..            ...                           ...                          ...   
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157         0.000                        517.0000                            1285.0
158         0.000                        233.9000                             996.0
159         0.028                        428.0360                            1328.0
160         0.038                       1540.0000                            1743.0
161         0.000                        309.0000                            1169.8

     Inclination [deg]  Transit Midpoint [days]  Transit Depth [%]  \
0               83.590              2457063.210           0.038467
1               88.390              2458330.351           0.249000
2               89.460              2458342.223           0.085000
3               86.500              2455321.724           2.571701
4               89.030              2459503.327           0.365000
..                 ...                      ...                ...
157             85.600              2456258.062           1.380000
158             84.890              2459252.981           0.081839
159             88.010              2454508.738           1.336300
160             83.450              2457917.475           0.909000
161             87.045              2458425.789           0.026759

     Transit Duration [hours]  Radial Velocity Amplitude [m/s]
0                    1.543900                            6.020
1                    3.240000                            3.800
2                    4.290000                            4.696
3                    2.387000                          590.000
4                    1.693000                            3.220
..                        ...                              ...
157                  2.426400                           62.000
158                  0.900451                            3.540
159                  2.683900                           90.170
160                  2.772000                           80.500
161                  2.952024                            1.185

[162 rows x 18 columns]
     Star ID  Planet ID
0          1          1
1          2          2
2          2          3
3          3          4
4          4          5
..       ...        ...
160      127        158
161      128        159
162      129        160
163      130        161
164      131        162

[165 rows x 2 columns]
    Discovery ID  Planet ID
0              1          1
1              2          2
2              3          2
3              2          3
4              3          3
..           ...        ...
64            52        152
65            53        155
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66            54        160
67            55        161
68            56        162

[69 rows x 2 columns]
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In [ ]: